

MATH – 241
Engineering Mathematics I
Problem Set – 4

Question 1. Let $\{v_1, v_2, \dots, v_n\}$ be a basis of a vector space V . Suppose W is a k -dimensional subspace of V , $1 < k < n$. Show that, for any subset $\{v_{i_1}, v_{i_2}, \dots, v_{i_m}\}$ of $\{v_1, v_2, \dots, v_n\}$ with $m > n - k$, there exists a nonzero vector $w \in W$ which is a linear combination of $\{v_{i_1}, v_{i_2}, \dots, v_{i_m}\}$.

Question 2. Let W_1 and W_2 be nontrivial subspaces of a vector space V . The sum $W_1 + W_2$ is called a **direct sum**, denoted as $W_1 \oplus W_2$, if every element $\alpha \in W_1 + W_2$ can be **uniquely** written as $\alpha = w_1 + w_2$ where $w_1 \in W_1$ and $w_2 \in W_2$.

Show that $M_2(\mathbb{R})$: the set of all 2×2 matrices with real entries is the direct sum $M_2(\mathbb{R}) = W_1 \oplus W_2$ where;

$$W_1 = \left\{ \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \mid a, b \in \mathbb{R} \right\},$$
$$W_2 = \left\{ \begin{bmatrix} c & d \\ d & -c \end{bmatrix} \mid c, d \in \mathbb{R} \right\}.$$

If V_1 is the subspace of $M_n(\mathbb{R})$ consisting of all $n \times n$ symmetric matrices, what will be a subspace V_2 such that $V_1 \oplus V_2 = M_n(\mathbb{R})$?

Question 3. Let $V = \left\{ x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 \mid x_1 = x_3 + x_4, x_2 = x_3 - x_4 \right\}.$

Find an orthonormal basis for V and for the subspace $V^\perp$ that contains all orthogonal vectors to V .

Question 4. If S is a **subset** of an inner product space V , answer true or false:

- (a) $S \cap S^\perp = \{0\}$.
- (b) $S \subseteq (S^\perp)^\perp$.
- (c) $(S^\perp)^\perp = S$.
- (d) $(S^\perp)^\perp = \text{Span}(S)$.
- (e) $[(S^\perp)^\perp]^\perp = S^\perp$.
- (f) $S^\perp$ is always a subspace of V .
- (g) $S^\perp \oplus \text{Span}(S) = V$.
- (h) $\dim S^\perp + \dim (S^\perp)^\perp = \dim V$.

Question 5. To find an orthonormal basis for $Span\{\underline{v}_1, \underline{v}_2, \dots, \underline{v}_m\}$ where $\{\underline{v}_1, \underline{v}_2, \dots, \underline{v}_m\}$ is a **linearly dependent** set of vectors, a procedure that is a case variant of the Gram-Schmidt procedure can be used. Illustrate this procedure on the following set of vectors:

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \\ 4 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Question 6. Let the matrix $\underline{\underline{A}}_{m \times n} = [\underline{a}_1, \dots, \underline{a}_n]$ have n linearly independent columns. By using Gram-Schmidt procedure show that $\underline{\underline{A}}$ can be decomposed as

$$\underline{\underline{A}} = \underline{\underline{Q}} \cdot \underline{\underline{R}} = [\underline{q}_1, \dots, \underline{q}_n] \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ 0 & r_{22} & \dots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & r_{nn} \end{bmatrix}$$

where $\underline{q}_1, \dots, \underline{q}_n$ are the orthonormal vectors generated by the G-S procedure and $\underline{\underline{R}}$ is an upper triangular matrix.

Question 7. Determine x_0 and v_0 from the available set of noise-free measurements

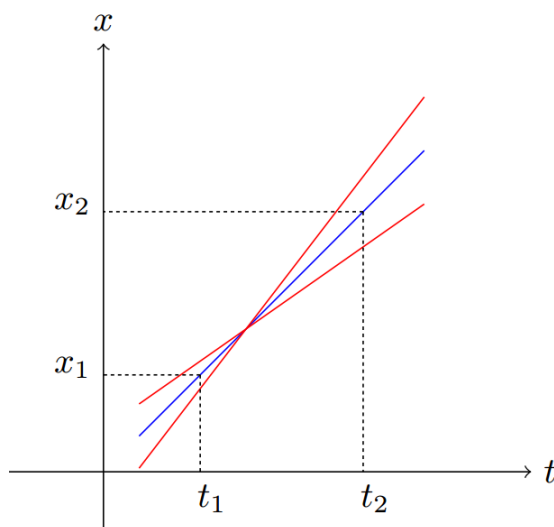
$$\{(t_1, x_1), (t_2, x_2)\} = \{(1, 15), (3, 25)\}.$$

In the above set of measurements we have ignored the presence of measurement noise. Assume that the available set of measurements are:

$$\{(t_1, \tilde{x}_1), (t_2, \tilde{x}_2)\} = \{(1, 16), (3, 22)\}.$$

Then determine x_0 and v_0 and compare your results with the noise-free case.

Now assume that we are given the following measurements: $\{(1, 16), (3, 22), (5, 39)\}$, determine the best approximation to these observations within the below kinematic model of $x = x_0 + t.v_0$.



$$x = x_0 + t.v_0$$